Mock Exam for the Computer Graphics Unit

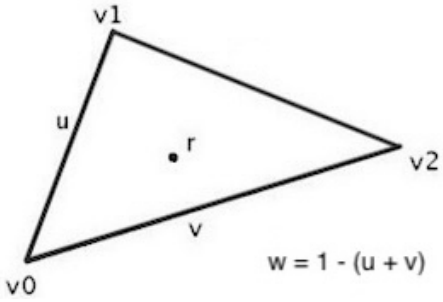
This year's Computer Graphics exam will be quite different from that of previous years. Past exams have required students to demonstrate a deep understanding of rendering algorithms by performing calculations on paper. The exam this year will not involve such calculations, but will instead assess higher-level understanding of the taught materials. This mock exam provides a sample set of questions to give students a feel for the kinds of questions that may be set. Please note that:

- Some questions in this mock paper are *relevant* questions from previous years;
- Some questions are *adapted* questions from previous years (that have been converted into multiple choice/answer format);
- Some questions are *unused candidate* questions from the current year's question pool.

This set of questions is just a sample of what to expect in the real exam. Note that some questions might *look* like they require some form of calculation, however these can be easily answered with knowledge of the principles of rendering (combined with deduction and disqualification of clear incorrect answers). Some questions may have more than one correct answer - be sure to mark all that apply. Some answers may be a combination of the available options.

Question 1: Barycentric Coordinates

Which one of the below calculations illustrate how barycentric coordinates are used to weight vertex values (for example when interpolating vertex brightnesses).



a. `v0*u ; v1*v ; v2*w`
b. `v0*v ; v1*w ; v2*u`
c. `v0*w ; v1*u ; v2*v`
d. `v0*v ; v1*u ; v2*w`
c. `v0*w ; v1*v ; v2*u`
[ANSWER]: (c)

Question 2: Rasterising

Which one of the following phenomenon would cause a skipped line (as shown in the diagram below) when drawing the filled grey triangle:



a. A gap between the "top" and "bottom" filled triangles
b. Incorrect interpolation of z depth
c. A problem with row number rounding
d. An overflow causing float wrap-around
e. A transparent row of pixels
[ANSWER]: (c)

Question 3: Casting Shadows

How can you determine if a point on a 3D surface is in shadow (mark all possible answers):
a. The camera's view of the point is obscured
b. The point's view of the light is obscured
c. The light's view of the point is obscured
d. The camera's view of the light is obscured
e. The point's view of the camera is obscured
[ANSWER]: (b) and (c)

Question 4: Specular Lighting

A render had a specular highlight with an exponent of 128, what will be the effect of changing the exponent to 256 (mark all possible answers):
a. the specular highlight will be larger
b. the specular highlight will be smaller
c. the specular highlight will appear higher
d. the specular highlight will appear lower
e. the specular highlight will remain unchanged
[ANSWER]: (b)

Question 5: Camera Orientation

A camera is at a position of `[58,42,30]` relative to the scene/world origin. We wish to orient the camera so that it points directly towards the position `[-12,61,-60]` with the camera in the vertical (i.e. with no 'roll' to its orientation). Which of the following is the one correct "forward" camera orientation vector:
a. `[-0.14, 0.71, -0.69]`
b. `[0.61, -0.16, 0.78]`
c. `[-0.61, 0.16, -0.78]`
d. `[70, -19, 90]`
e. `[-70, 19, -90]`

[ANSWER]: (a) incorrect - just normalised LookAt position (b) correct ; (c) incorrect - inverted direction ; (d) incorrect - unnormalised ; (e) incorrect - unnormalised inverted direction

Question 6: OBJ File Content

Answer the following questions regarding the content and formatting of OBJ files by marking one option:

i) What token indicates a line containing the [x,y,z] coordinates of a vertex (mark one answer) ?
a. f
b. v
c. o
d. Kd
e. vn
[ANSWER]: (b)

ii) What token indicates a line that contains the name of a 3D object (mark one answer) ?
a. f
b. v
c. o
d. Kd
e. vn
[ANSWER]: (c)

iii) What marker indicates a line containing the RGB channels of a material colour (mark one answer) ?
a. f
b. v
c. o
d. Kd
e. vn
[ANSWER]: (d)

iv) What is the numerical range for each RGB colour channel (mark one answer) ?
a. -1.0 to 1.0
b. 0.0 to 1.0
c. 0 to 255
d. 1 to 256
e. 0 to 1024
[ANSWER]: (b)

Question 7: Closest Intersection

You may remember the 'closest intersection' code fragment from the unit workbooks:

```
glm::vec3 e0 = triangle.vertices[1] - triangle.vertices[0];
glm::vec3 e1 = triangle.vertices[2] - triangle.vertices[0];
glm::vec3 SPVector = cameraPosition - triangle.vertices[0];
glm::mat3 DEMatrix(-rayDirection, e0, e1);
glm::vec3 possibleSolution = glm::inverse(DEMatrix) * SPVector;
```

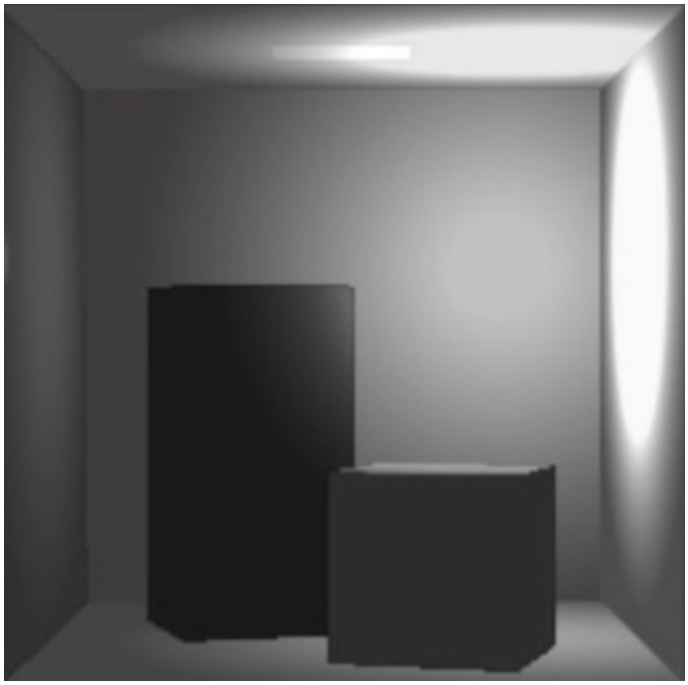
For a particular 3D model, the following five 'possible solutions' were identified by the above algorithm. Indicate which one of these would be the correct 'closest intersection' for use in generating a ray traced view of a scene:

a. `[0.21,0.62,0.78]`
b. `[0.77,0.38,0.34]`
c. `[0.33,-0.26,0.31]`
d. `[0.64,0.31,0.28]`
e. `[-0.24,0.28,0.26]`

[ANSWER]: (a) off triangle; (b) not nearest; (c) off triangle; (d) correct ! ; (e) behind camera

Question 8: Rendering Problems

Which rendering errors would cause the noticeable rendering issue illustrated in the image below (mark all possible correct answers):

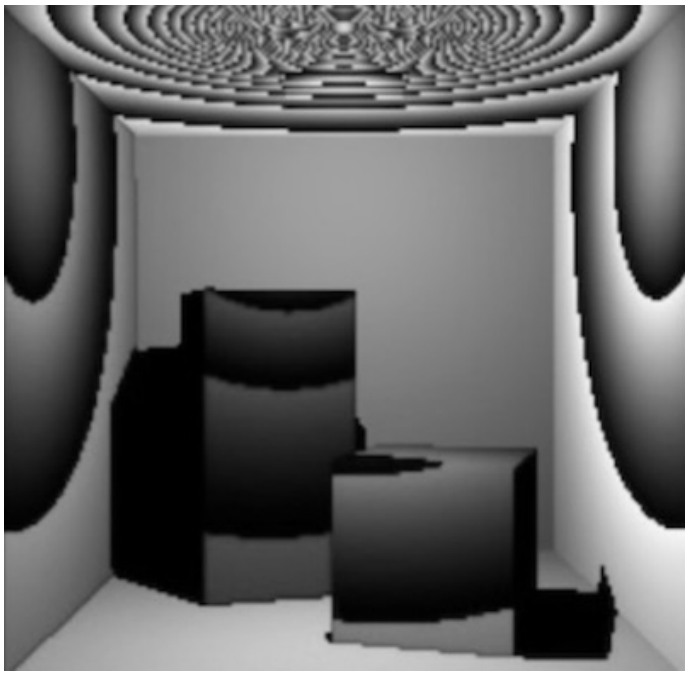


a. The light has been positioned in an inappropriate location
b. Two separate lights are being used in different location
c. Proximity and AoI lighting have not be weighted appropriately
d. Vertex normals have not been correctly calculated
e. Barycentric interpolation has not been correctly applied

[ANSWER]: (c)

Question 9: More Rendering Problems

Which rendering errors would cause the noticeable rendering issue illustrated in the image below (mark all possible correct answers):



a. An incorrect model scaling has been used
b. Brightness values are underflowing (wrapping around)
c. Shadow rays are intersecting with their own triangles
d. The "forward" vector is inverted (pointing in wrong direction)
e. Z depth has been used instead of 1/Z
[ANSWER]: (b)

Question 10: Traingle Splitting

Why is it beneficial to split a triangle in two when drawing filled triangles (mark all possible correct answers):
a) It makes it easier to calculate the start and end points of rake rows
b) It makes it easier to calculate the number of rows in the triangle
c) It makes it easier to calculate the z-depth of each point on the triangle
d) It is more efficient to render since there are fewer pixels to draw
e) It is more efficient for rendering flat-bottomed triangles
ANSWER: (a)

Question 11: Peek Through

What can cause the two cubes to peek through the floor of the Cornell to the extent shown below (mark all possible correct answers):



a) The z-depth of floor being interpolated linearly
b) Unsuitable rounding of the vertex positions
c) Use of an uninitialised depth buffer
d) A fault in the rotation about the X axis
e) Use of an unsuitable focal length
ANSWER: (a)

Question 12: Model Size

What can cause the "bunnyzilla" problem that was described in the workbooks (mark all possible correct answers):
a) Applying too large an image-plane scaling
b) Inappropriate positioning of camera resulting in strange perspective
c) The image plane being positioned too close to the world origin
d) Rendering two models created with different model scaling
e) A radiation leak at the rabbit farm
ANSWER: (d)

Question 13: Look At

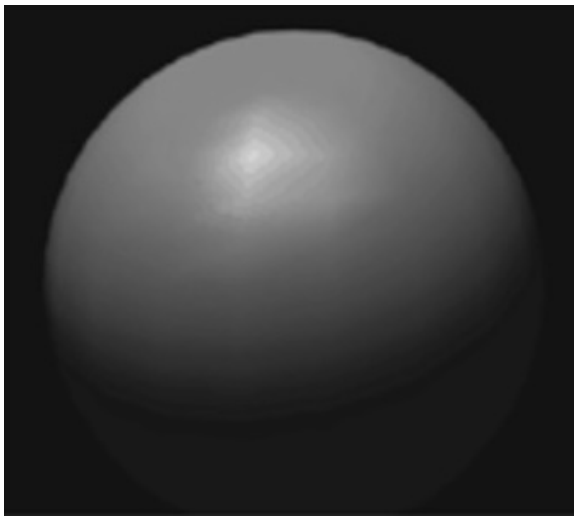
How does the "lookAt" function effect the camera (mark all correct effects)?
a) Rotates the camera position
b) Rotates the camera orientation
c) Alters the focal length
d) Translates the camera position
e) Translates the camera orientation
ANSWER: (b)

Question 14: Lighting

Which one the following types of lighting would be most effective in making a surface appear shiny/glossy ?
a) Ambient lighting with a high threshold
b) Diffuse lighting with an oblique angle of incidence
c) Diffuse lighting with an acute angle of incidence
d) Specular lighting with a high exponent
e) Specular lighting with a low exponent
ANSWER: (d)

Question 15: Sphere Rendering

Which type of shading would produce the below render of a sphere:



a) Flat
b) Gouraud
c) Phong
d) Barycentric
e) Occlusion
ANSWER: (b)